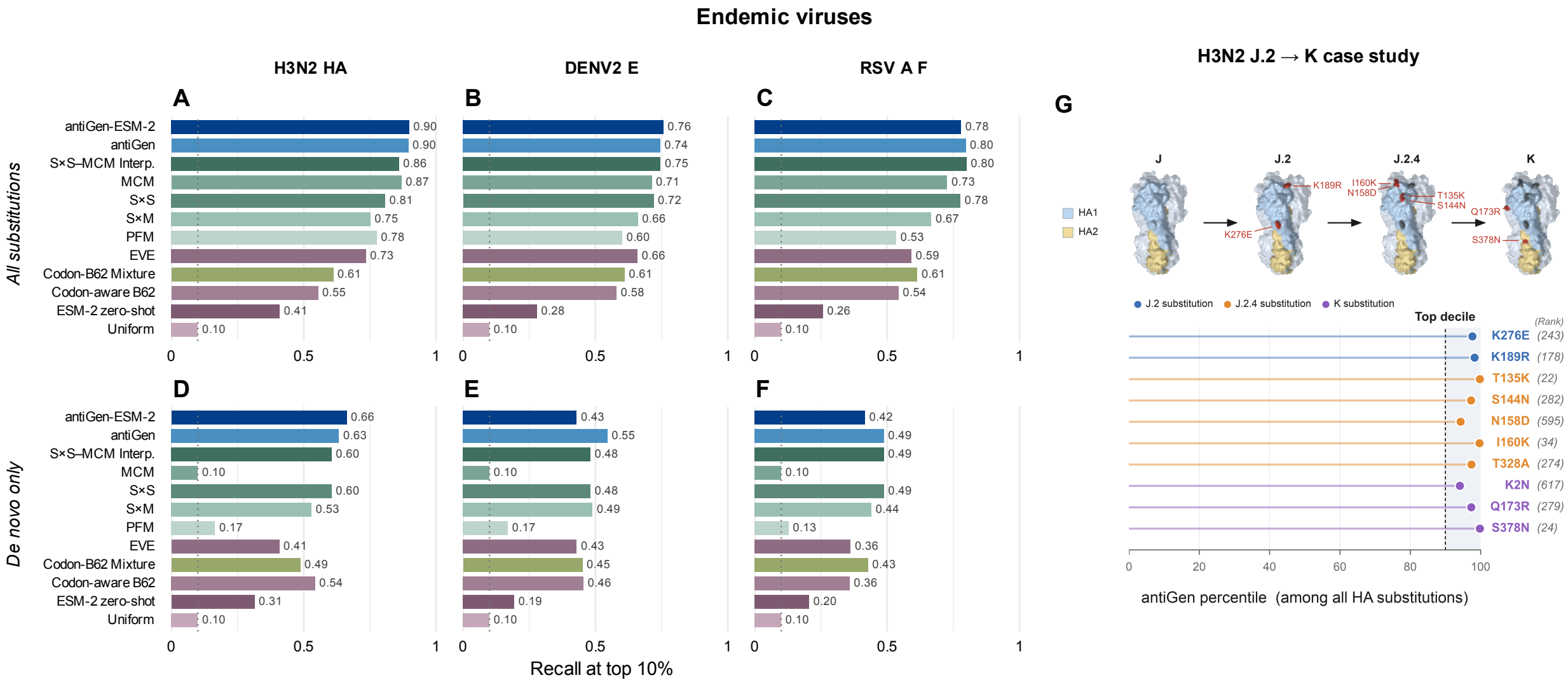

Endemic viruses
H3N2 HA
DENV2 E
RSV A F
A
B
C
D
E
F
All substitutions
De novo only
antiGen-ESM-2 0.90 0.76 0.78 0.66 0.43 0.42
antiGen 0.90 0.74 0.80 0.63 0.55 0.49
S×S–MCM Interp. 0.86 0.75 0.80 0.60 0.48 0.49
MCM 0.87 0.71 0.73 0.10 0.10 0.10
S×S 0.81 0.72 0.78 0.60 0.48 0.49
S×M 0.75 0.66 0.67 0.53 0.49 0.44
PFM 0.78 0.60 0.53 0.17 0.17 0.13
EVE 0.73 0.66 0.59 0.41 0.43 0.36
Codon-B62 Mixture 0.61 0.61 0.61 0.49 0.45 0.43
Codon-aware B62 0.55 0.58 0.54 0.54 0.46 0.36
ESM-2 zero-shot 0.41 0.28 0.26 0.31 0.19 0.20
Uniform 0.10 0.10 0.10 0.10 0.10 0.10
0 0.5 1
Recall at top 10%
G
H3N2 J.2 → K case study
HA1
HA2
J
J.2
J.2.4
K
K189R
I160K
N158D
T135K
S144N
Q173R
K276E
S378N
J.2 substitution
J.2.4 substitution
K substitution
Top decile
(Rank)
K276E (243)
K189R (178)
T135K (22)
S144N (282)
N158D (595)
I160K (34)
T328A (274)
K2N (617)
Q173R (279)
S378N (24)
0 20 40 60 80 100
antiGen percentile (among all HA substitutions)